NG JING YAO

Software Engineering Graduate | Full-Stack Development | AI Application Development

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PROFILE SUMMARY

Software Engineering graduate with experience in building full-stack applications and AI-powered applications. Experienced in Python, Flask, React.js, API integration, Docker, Retrieval-Augmented Generation (RAG), document retrieval, LLM APIs, LLM tool calling, and prompt engineering.

EXPERIENCE

Intern

YTL Construction Sdn Bhd | Jan 2026 - Jul 2026

- Developed and piloted a full-stack internal RAG chatbot for employees to retrieve IT and HR knowledge through a SharePoint-integrated chat interface.
- Built the frontend using SharePoint Framework (SPFx), React.js, and TypeScript, supporting streaming responses, conversation memory, feedback collection, and error reporting.
- Collaborated with approximately 20 internal users to improve retrieval quality, knowledge coverage, answer reliability, and feedback workflows.

Software Development Intern

YYC GST Consultants Sdn Bhd | Nov 2023 - Feb 2024

- Assisted in developing web application modules using Vue.js and Laravel, contributing to user interface improvements and application functionality.
- Collaborated with the development team using GitHub for version control and front-end/back-end integration.

PROJECTS

Full-Stack RAG Chatbot

YTL Construction

- Built a full-stack RAG chatbot over 20+ internal IT and HR knowledge documents using Flask, LangChain, ILMU LLM APIs, and a hybrid retrieval pipeline with ChromaDB vector search, BM25 keyword search, and cross-encoder reranking.
- Developed a document ingestion pipeline for Markdown/PDF knowledge sources, including hierarchical chunking, metadata tracking, abbreviation extraction, vector embedding generation, and incremental re-indexing.
- Implemented query preprocessing for conversation-aware query rewriting, language translation, abbreviation expansion, and multi-query generation to improve retrieval quality.
- Implemented 5 LLM tool-calling workflows with backend validation, human-in-the-loop confirmation, and guardrails for source-grounded answers and sensitive information.

EasyHealthy

Final Year Project (FYP)

- Developed a mobile diet and fitness tracker application using Flutter and Firebase, featuring meal logging, custom foods, recipe recommendations, and USDA food data integration.
- Implemented a hybrid food recommendation system, progress charts, and user authentication using Firebase Authentication and Cloud Firestore.

TECHNICAL SKILLS

Backend: Python, Flask, Java, PHP, Laravel, REST APIs
Frontend: React.js, TypeScript, JavaScript, Vue.js, Flutter, Dart, HTML, CSS
Databases: SQL, MySQL, Firebase
Deployment: Docker, Linux, Internet Information Services (IIS)
AI/LLM: Retrieval-Augmented Generation (RAG), prompt engineering, LLM APIs, LLM tool/function calling, LangChain, ChromaDB, BM25, cross-encoder reranking
Tools: Git, GitHub, XAMPP, Codex, Claude Code

EDUCATION

Bachelor of Software Engineering

Tunku Abdul Rahman University of Management and Technology (TARUMT) | Jul 2024 - Aug 2026

CGPA: 3.73 | Dean's List: Nov 2024 | President’s List: Jan 2026

Diploma in Information Technology

Tunku Abdul Rahman University of Management and Technology (TARUMT) | Jun 2022 - Jun 2024

CGPA: 3.79

LANGUAGES

English (Proficient) | Chinese (Native) | Malay (Intermediate)